The Top 25 Laboratory Tests by Volume and Revenue in Five Different Countries

Susan Horton, PhD,[1] *Kenneth A. Fleming, MBChB,*[2] *Modupe Kuti, MBBS,*[3] *Lai-Meng Looi, MBBS,*[4] *Sanjay A. Pai, MD,*[5] *Shahin Sayed, MBChB, MMed,*[6] *and Michael L. Wilson, MD*[7]

From the [1]School of Public Health and Health Systems, University of Waterloo, Waterloo, Canada; [2]Green Templeton College, University of Oxford, Oxford, UK; [3]Chemical Pathology, College of Medicine, University of Ibadan, Ibadan, Nigeria; [4]Pathology, University of Malaya, Kuala Lumpur, Malaysia; [5]Columbia Asia Referral Hospital, Bangalore, India; [6]Pathology, Aga Khan University, Nairobi, Kenya; and [7]Pathology and Laboratory Services, Denver Health Medical Center, Denver, CO.

Key Words: Economic evaluation; Global health; Test volume; Revenue; India; Kenya; Malaysia; Nigeria; United States

Am J Clin Pathol May 2019;151:446-451

DOI: 10.1093/AJCP/AQY165

Downloaded from academic.oup.com/ajcp/article/151/5/446/5237639 by University of Calgary user on 15 October 2026

ABSTRACT

Objectives: *To compare the most common diagnostic/laboratory tests across five different referral hospitals by volume and revenue.*

Methods: *The authors obtained data on volumes and reimbursement rates for the most common 25 tests at the five hospitals with which they are affiliated and organized them to be as comparable as possible. Simple descriptive statistics were used to make cross-country comparisons.*

Results: *There are strong similarities across all five hospitals in the top five tests by both volume and revenue. However, the top five by volume differ from the top five by revenue. Reimbursement rates also follow common patterns, being lowest for the most common biochemical test; intermediate for the most common hematology and microbiology tests, respectively; and highest for the most common pathology test.*

Conclusions: *Most of the most common tests also appear in the new Essential Diagnostics List. This may inform plans for universal health coverage.*

Access to, and affordability of, laboratory tests is key to attaining quality medical care. This is particularly important in low- and middle-income countries (LMICs) where public financing of care is limited and out-of-pocket payment for diagnostic tests is often the norm.[1] Economic information about diagnostic costs will be increasingly necessary as governments are urged to plan for universal health coverage, Sustainable Development Goal 3.8.[2]

The objective of the present study was to examine comparative volumes and revenues of commonly performed tests across a range of countries, in a standard setting—namely, the laboratories of large referral hospitals. This builds on earlier work for the same hospitals that identified the most common tests.[1]

The cross-country comparisons represent a contribution to the literature, comparing hospitals in four LMICs and one high-income country. This can provide information as to how far per capita income is associated with trends in test use, as well as help to identify unique tests related to particular local epidemiology. This may be useful with the issuance of an Essential Diagnostics List (EDL) by the World Health Organization[3] and the need for individual countries to develop national EDLs.

Volume of tests run is one indicator of the importance of particular tests. Some tests are inexpensive to run and have to be rerun frequently for the same patients to monitor their status (eg, blood glucose). Other tests are run infrequently but require considerably more time and resources and may be run very few times for the same

© American Society for Clinical Pathology, 2018. All rights reserved.
For permissions, please e-mail: journals.permissions@oup.com

patient but have a major impact on treatment, such as histopathology tests for cancer. If we assume price (or reimbursement rate) per test is an indication of time and effort required to run the test, then ranking tests by revenue can also provide important information, and the ranking is likely to differ from that by volume.

Materials and Methods

Information was obtained on the number of laboratory tests run in 2015 (four of the hospitals) and 2016 (Aga Khan University Hospital [AKUH]) in the five hospitals with which five authors (M.K., L.L., S.P., S.S., and M.W.) are associated. These hospitals included three in lower-middle-income countries, one in an upper-middle-income country, and one in a high-income country. Two of these are private hospitals and the other three are public. The AKUH in Nairobi, Kenya, is a private hospital with 254 beds; Columbia Asia Referral Hospital (CA) in Bangalore, India, is also a private hospital with 160 beds; University College Hospital (UCH) in Ibadan, Nigeria, is a public hospital with 850 beds; University of Malaya Medical Centre (UMMC) in Kuala Lumpur, Malaysia, is a public hospital with 1,060 beds; and Denver Health in Denver, Colorado, is a public safety net hospital with 525 beds. Point-of-care tests were not included.

Data were obtained from hospital records on all tests run for inpatients and outpatients visiting the hospital, for the 25 most common tests, as well as the standard price charged by the hospital (for private hospitals that set the test price) or standard reimbursement rate (for public hospitals, where the public sector sets the reimbursement rate). The prices of tests in public hospitals are generally set by governments and are not necessarily perfectly correlated with the actual cost of running the test.

Tests are grouped somewhat differently in different laboratories. In some cases, hospitals record similar tests under separate codes, and we used best judgement to group tests appropriately for comparison. For example, fasting, postprandial, and random blood glucose were combined as a group of all blood glucose tests, and we calculated a weighted average price/reimbursement rate for this combined test. In other cases, hospitals (particularly those in higher-income countries) run panels of tests rather than individual tests. As an example, a renal function test might involve as few as two tests (serum urea nitrogen and creatinine), whereas at Denver Health, it is most often combined with another six tests (three electrolytes—namely, sodium, potassium, and chloride—as well as glucose, carbon dioxide, and calcium) as part of a "basic metabolic panel." Some hospitals use different tests for the same condition; for example, Denver Health most frequently uses the enzyme immunoassay method for syphilis screening, whereas UMMC uses the *Treponema pallidum* hemagglutination method. These differences mean that we came up with a broad overview of the common tests and costs, leaving more detailed analysis for future research.

The price or reimbursement rate for hospitals is not always public information. We were given permission to present rankings of tests by value and relative but not actual prices/reimbursement rates for some of the hospitals. We accordingly calculated the relative prices of the most common hematology test at each hospital (CBC), as well as the most common microbiology test (urine culture) and histopathology tests (surgicals), to the price of the most common biochemical test (blood glucose) at the same hospital.

Results

Table 1 provides information on the most common 25 tests by volume. In all the tables, the hospitals are ranked left to right in order of increasing per capita income in the country, measured using market exchange rates.[4] The top five tests by volume show strong similarities across all five hospitals. A CBC is the most common test in four of the five hospitals (and is the second most common in the fifth hospital). Tests of renal function were among the top five in all five hospitals (described as "renal function tests" at CA and UMMC; as "urea, electrolytes" at AKUH; and as "electrolytes, urea, and creatinine" at UCH; the "basic metabolic panel" at Denver Health contains seven of the 10 elements of the usual US CPT80069 renal function test). A (blood) glucose test is also one of the top five tests by volume in all the hospitals. Three of the five hospitals also include urinalysis in the top five tests. Only one (or, in one case, two) of the top five tests by volume in each hospital was significantly different from that of the other hospitals. This may reflect varying local clinical practice patterns/epidemiology. These local variations include stool microscopy (AKUH), thyroid function tests (CA), blood group and crossmatch (UCH), liver function tests and serum magnesium (UMMC), and *Chlamydia* detection (Denver Health).

The top five tests by volume at UMMC account for 46.5% of all laboratory tests and 37.5% at Denver Health (comparable data were not available for the other hospitals). The top 25 similarly account for 79% of all tests at UMMC and 71% at Denver.[2]

The top five tests by revenue are somewhat different Table 2. The CBC still appears in this list for all five hospitals, although at the top of the list only in two hospitals, and the tests of renal function also remain in the top five. The liver function tests now appear in the top five for four of the hospitals, thyroid function in three, and blood group

Downloaded from academic.oup.com/ajcp/article/151/5/446/5237639 by University of Calgary user on 15 October 2026

© American Society for Clinical Pathology

DOI: 10.1093/ajcp/aqy165

Table 1
Top 25 Tests by Volume at Five Referral Hospitals

Rank	AKUH, Nairobi	CA, Bangalore	UCH, Ibadan	UMMC, Kuala Lumpur	Denver Health
1	CBC	CBC[a]	CBC	CBC	CBC
2	Urinalysis	Glucose	Electrolytes, urea, and creatinine	Renal function	Basic metabolic panel[b]
3	Urea, electrolytes	Creatinine	Glucose	Liver function	Glucose
4	Stool microscopy	TSH	Urinalysis	Blood glucose	Urinalysis
5	Glucose	Thyroid function	Blood group and crossmatch	Magnesium serum	*Chlamydia* detection
6	C-reactive protein	Renal function	Liver function	Lipid profile	HbA_{1C}
7	Malarial parasites	Urinalysis	Lipid profile	PT/INR	Phosphorus
8	Stool *Helicobacter pylori* antigen	Potassium	Blood film for malaria parasite	APTT	Magnesium
9	Liver function	Liver function	Urine microscopy	Urinalysis	PT/INR
10	Urine microscopy	Platelet count	AFB studies	HbA_{1C}	Comprehensive metabolic panel[c]
11	Surgicals[d]	Urine culture	PT/INR	Blood group	TSH
12	Crossmatch	Lipid profile	Other microscopy	Calcium, phosphorus	Liver function
13	Malaria antigen	PCV	Surgicals	Blood culture[e]	Lipid panel[f]
14	HbA_{1C}	HbA_{1C}	Hb electrophoresis	HBsAg	PTT
15	Lipid profile	Vitamin B_{12}	Cytology	HIV combo	Troponin-I
16	HIV	Hemoglobin	Syphilis	HCV	Urine culture
17	Thyroid function	Sodium	Blood culture	C-reactive protein	Blood group
18	ESR	Vitamin D	ESR	CKMB	Lactate
19	TSH	Blood group	HIV	ESR	Antibody screen
20	Unit packed RBCs	Calcium	Blood group	Thyroid function	Drugs of abuse screen, urine
21	Pregnancy	Coagulation profile	HCV	Syphilis	Cytology
22	Cytology	CBC with ESR	Stool microscopy	Troponin	Syphilis
23	Calcium	Blood culture	HBsAg	Uric acid serum	Surgicals
24	Blood culture	Electrolytes	HbA_{1C}	Lactate	Blood culture
25	Syphilis	ALT	Phosphorus	Urine culture	Pregnancy

AFB, acid-fast bacilli; AKUH, Aga Khan University Hospital; ALT, alanine aminotransferase; APTT, activated partial thromboplastin time; CA, Columbia Asia Referral Hospital; CKMB, creatine kinase muscle/brain; ESR, erythrocyte sedimentation rate; Hb, hemoglobin; HbA_{1C}, hemoglobin A_{1C}; HBsAg, hepatitis B surface antigen; HCV, hepatitis C; HIV, human immunodeficiency virus; PCV, packed cell volume; PT/INR, prothrombin time/international normalized ratio; PTT, partial thromboplastin time; TSH, thyroid-stimulating hormone; UCH, University College Hospital; UMMC, University of Malaya Medical Centre.
[a]Includes CBC plus smear (hemogram).
[b]Includes calcium, serum urea nitrogen, creatinine, carbon dioxide, potassium, chloride, and sodium (ie, contains renal function tests plus others).
[c]Includes calcium, serum urea nitrogen, creatinine, carbon dioxide, potassium, chloride, and sodium (ie, contains renal function tests plus others), plus gap-calculated ALT, alkaline phosphatase, total protein, albumin, and total bilirubin.
[d]Surgicals refers to all histopathology tests.
[e]Includes both aerobic and anaerobic culture.
[f]Includes cholesterol, triglyceride, high-density lipoprotein, and calculated low-density lipoprotein.

in two. As with volume of testing, each hospital has either one or two unique tests in the top five by revenue. These are urinalysis (AKUH), vitamin D (CA), lipids (UCH), blood culture (UMMC), and *Chlamydia* detection as well as urine screening for drugs of abuse (Denver Health).

The reason for the changes in ranking is obviously related to the price of individual tests. Liver function and thyroid function are in the middle or highest price category per test, and the unique tests generally appear in the highest price categories at the respective hospitals. One striking difference between the rankings by volume and revenue is that surgicals (ie, histopathology tests) appear in the top 10 list for three of the hospitals by revenue but much further down the same list by volume. This is explained by the much greater resource requirements for this test compared with others, as illustrated below.

Table 3 presents relative prices for different types of tests within each hospital, comparing the most common biochemical test (blood glucose), the most common hematologic test (CBC), the most common microbiology test (urine culture), and histopathologic tests overall (surgicals). The biochemical test price is lowest in each hospital, the hematology and microbiology test prices are about three times as high, and the histopathology test price is about 15 times as high as the price of the glucose test, using the median relative price of the five hospitals (Table 3). Even though the relative prices vary somewhat across hospitals, the biochemistry test price is always lowest, with the hematology and microbiology test prices being intermediate and the histopathologic test price being the highest.

Discussion

Despite the differences in health systems, local epidemiology, practice patterns, and resources, there are strong

Downloaded from academic.oup.com/ajcp/article/151/5/446/5237639 by University of Calgary user on 15 October 2026

DOI: 10.1093/ajcp/aqy165
© American Society for Clinical Pathology

Table 2
Top 25 Tests by Revenue at Five Hospitals[a]

Rank	AKUH, Nairobi	CA, Bangalore	UCH, Ibadan	UMMC, Kuala Lumpur	Denver Health
1	CBC	Renal function	Electrolytes, urea, and creatinine	Renal function	*Chlamydia* detection
2	Stool *Helicobacter pylori* antigen	Thyroid function	CBC	CBC	Basic metabolic panel
3	Urea, electrolytes	Liver function	Blood group and crossmatch	Blood culture	CBC
4	Liver function	CBC	Liver function	Liver function	Drugs of abuse screen, urine
5	Thyroid function	Vitamin D	Lipid profile	Blood group	TSH
6	Urinalysis	Vitamin B_{12}	Other microscopy	HbA_{1C}	Surgicals
7	Surgicals	Lipid profile	Surgicals	Lipid profile	HbA_{1C}
8	C-reactive protein	TSH	Blood culture	Troponin	Glucose
9	Urine microscopy, culture, and sensitivity	HbA_{1C}	Urine microscopy	Hepatitis C	Comprehensive metabolic panel
10	Lipid profile	Urine culture	Urinalysis	C-reactive protein	Lipid panel
11	HbA_{1C}	Blood culture	PT/INR	Thyroid function	Magnesium
12	Glucose	Creatinine	Glucose	Urine culture	Phosphorus
13	Blood culture	Coagulation profile	Hb electrophoresis	PT/INR	Liver function
14	Unit packed RBCs	Urinalysis	Syphilis	HIV combo	Urinalysis
15	Stool microscopy	Potassium	HCV	APTT	Urine culture
16	HIV	Glucose	Cytology	CKMB	PT/INR
17	ESR	CBC with ESR	HBsAg	Glucose	Troponin-I
18	TSH	Electrolytes	HbA_{1C}	Urinalysis	PTT
19	Malarial parasites	Blood group	Blood film for malaria parasite	HBsAg	Blood culture
20	Crossmatch	Calcium	Stool microscopy	Calcium phosphorous	Cytology
21	Cytology	Platelet count	Blood group	Magnesium serum	Blood group
22	Pregnancy	Sodium	ESR	Syphilis	Antibody screen
23	Calcium	PCV	Phosphorous	Lactate	Syphilis
24	Malaria antigen	ALT	AFB studies[b]	Uric acid serum	Lactate
25	Syphilis	Hemoglobin	HIV[b]	ESR	Pregnancy

[a]Note that these are the rankings of the top 25 tests by revenue, restricted to those tests that all appear in the top 25 by volume. It is possible that there are other less common (but very expensive tests) that would appear in the top 25 tests by revenue overall, but we were not able to obtain those data. For definitions of abbreviations, see Table 1.
[b]These two tests are provided free at UCH in Ibadan (through a President's Emergency Plan for AIDS Relief clinic).

Table 3
Relative Price of the Most Common Biochemistry, Hematology, Microbiology, and Histopathology Tests at Five Referral Hospitals[a]

Test Type	AKUH, Nairobi	CA, Bangalore	UCH, Ibadan	UMMC, Kuala Lumpur	Denver Health
Biochemistry	1.00	1.00	1.00	1.00	1.00
Hematology	1.58	2.96	1.33	6.67	1.89
Microbiology	3.38	6.76	3.17	7.50	2.55
Histopathology	6.36	15.03	10.33	38.10	13.56

AKUH, Aga Khan University Hospital; CA, Columbia Asia Referral Hospital; UCH, University College Hospital; UMMC, University of Malaya Medical Centre.
[a]For this comparison, the relative price of the most common hematology test in our sample (CBC), the price of the most common microbiology test (urine culture), and the price per histopathology test (surgicals) were indexed to the price of the most common biochemistry test (blood glucose) at the same hospital.

similarities in laboratory test utilization patterns across five hospitals in countries with very different levels of per capita income. Four of the top five tests by volume are the same across four of the hospitals (and the fifth hospital has three that are the same). This is consistent with results from a hematology and clinical chemistry central laboratory in a hospital in Thailand,[5] which found that the five most common tests (in descending order) were CBC, creatinine, serum urea nitrogen, glucose, and urinalysis; for a smaller hospital clinical biochemistry laboratory in Zimbabwe[6] where the same list was urea, creatinine, sodium, potassium, and glucose; and for all private laboratories in South Africa[7] where the list was platelet count, CBC, creatinine, a combined test (sodium, potassium, chloride, carbon dioxide, and urea), and finally glucose.

The unique tests in the top five in the five hospitals we studied reflect local clinical practice patterns, specialization, or disease epidemiology. These tests are stool

Downloaded from academic.oup.com/ajcp/article/151/5/446/5237639 by University of Calgary user on 15 October 2026

© American Society for Clinical Pathology

DOI: 10.1093/ajcp/aqy165

microscopy at AKUH (possibly related to high levels of tropical infections), thyroid function at CA (a somewhat more complex test not readily available in many smaller laboratories in India, which is often referred to this private hospital), blood group and crossmatch at Ibadan (which provides a wide array of specialist surgical services for its locale), liver function tests and magnesium at UMMC, and finally *Chlamydia* detection at Denver Health (where the main hospital laboratory also serves as the Denver Public Health Laboratory and therefore supports testing related to sexually transmitted infections).

Although we have focused here only on the top five tests, previous research has shown that tests ranked sixth to 25th by volume also have strong similarities across these five hospitals.[2]

The large share of the top five tests in total test volume at UMMC (46.5%) and Denver Health (37.5%) is similar to that in a large hospital central laboratory in Thailand[5] (35.6%) and in the top 50 tests run by private laboratories in South Africa[7] (24.6%). Likewise, the share of the top 25 tests in total test volume at UMMC (79%) and Denver Health (71%) is consistent with similar large shares in Thailand[5] (95.5%) and South Africa[7] (61.3%).

The ranking of tests of different types by price bears a strong similarity to findings for the United Kingdom for test costs,[8] which found that a common (unspecified) hematology test cost 2.4 times as much as a common (unspecified) biochemistry test, while a standard microbiology test cost 6.9 times as much and a standard histopathology test 48.1 times as much. These relative prices/costs tend to be driven by test complexity, volume, and the amount of skilled labor required. Neither our data nor the UK study[8] included the new molecular tests (such as BRCA gene tests) that form only a small proportion of test volume but can can be reimbursed at 120 times as much as a glucose test (data for Denver Health).

Our list of the top five tests by revenue (CBC, renal function tests, thyroid function tests, liver tests, blood group, and another unique to the hospital) is also similar to findings for South Africa,[7] where the top five tests by revenue are creatinine (a component of renal function tests), CBC, cholesterol, thyroid function, and miscellaneous microscopy of bodily fluids.

In May 2018, the World Health Organization (WHO) published its first edition of the Model List of Essential In Vitro Diagnostics.[3] This is a group of tests for use at various levels of a tiered national health care system, including both primary care facilities with no or limited laboratory capacity and facilities with clinical laboratories. It is divided into two sets: a group of general laboratory tests for a wide array of communicable and noncommunicable diseases (58 tests) and a group for the WHO key disease areas: human immunodeficiency virus, tuberculosis (TB), malaria, hepatitis B virus/hepatitis C virus, human papillomavirus, and syphilis (55 tests).

Looking at this list in comparison to the list of top 25 tests identified in our survey, it is perhaps not surprising, if reassuring, that virtually all the tests are found in the WHO list. However, there is one apparent surprise. Given the importance of TB globally, the fact that there is only one mention of TB diagnosis in the top 25 tests in our five hospitals—acid-fast bacilli studies at UCH in Ibadan—is striking. There are several possible explanations, but the likeliest is that the tests are being done outside the hospital system, in, for example, charitable or governmental facilities dedicated to TB.

Of note, a clear message from both our top 25 list and the WHO EDL is that the most important (as measured by usage) tests globally are the relatively simple, well-tried, and long-established tests and that efforts to increase provision of pathology and laboratory medicine in countries with inadequate capacity should focus on providing such tests.

A limitation of our study is that we have used a convenience sample of five referral hospitals. It would be desirable to have a larger selection of health care institutions at different levels for each country (similar to the comprehensive data for Indiana from 15 hospitals and over a hundred clinics[9]). While we have used expert judgment to deal with issues of differences in test names and grouping of test components, some issues of interoperability may remain.

Another limitation is that we do not have coverage of point-of-care tests that are growing in volume. Although these constitute a relatively small proportion of test volume at referral hospitals (6.05% at Denver Health in 2017 and less than that in each of the other four hospitals considered here), they are more important in primary care settings in high-income countries, particularly the United States and the United Kingdom.[10,11] In LMICs, there has been considerable effort by organizations such as the Global Fund to make point-of-care tests for the major infectious diseases available at primary health centers.[12] Point-of-care tests account for 27% of the dollar value of medical diagnostic tests globally (authors' calculation using industry figures[12,13]). However, given that point-of-care tests generally cost more than laboratory tests,[12] this represents a correspondingly smaller share of the volume of tests globally.

Downloaded from academic.oup.com/ajcp/article/151/5/446/5237639 by University of Calgary user on 15 October 2026

DOI: 10.1093/ajcp/aqy165
© American Society for Clinical Pathology

Conclusions

As efforts to implement universal health coverage accelerate and expand, health care systems will need better access to diagnostic laboratory medicine services to guide diagnosis and treatment. The EDL is one way to help to prioritize resources, as part of ongoing efforts to build an appropriately resourced laboratory system,[14,15] guided by a national laboratory plan. As part of such a plan, it will be critical for governments to understand better which tests are used most commonly in the different tiers of health care systems as well as their relative cost. As shown in this article, the most common laboratory tests used in a geographically diverse set of referral hospitals are quite similar, and they also feature in the WHO EDL. More data should be collected from a more representative set of health facilities in each country, and this will likely be useful in developing national EDLs. The challenge remains to ensure that these tests will be affordable.

Corresponding author: Susan Horton, PhD; sehorton@uwaterloo.ca.

References

1. Horton S, Sullivan R, Flanigan J, et al. Delivering modern, high-quality, affordable pathology and laboratory medicine to low-income and middle-income countries: a call to action. *Lancet.* 2018;391:1953-1964.
2. World Health Organization. SDG 3: ensure healthy lives and promote wellbeing for all at all ages. http://www.who.int/sdg/targets/en/. Accessed August 14, 2018.
3. World Health Organization. Model list of essential in vitro diagnostics. 2018. www.who.int/medical_devices/diagnostics/WHO_EDL_2018.pdf. Accessed July 23, 2018.
4. World Bank. World Bank open data. https://data.worldbank.org. Accessed August 13, 2018.
5. Charuruks N, Chamnanpai S, Seublinvog T. Cost analysis of laboratory tests: a study of the central laboratory of King Chulalongkorn Memorial Hospital. *J Med Assoc Thai.* 2004;87:955-963.
6. Makuwaza L, Musarurwa C, Gomo Z. Evaluation of cost per test of clinical biochemistry tests at Parirenyatwa Central Hospital Laboratory, Harare, Zimbabwe. *Cent Afr J Med.* 2009;55:59-63.
7. Pretorius C. Utilisation of pathology procedures in the South African private pathology sector between 2003 and 2005. *S Afr Med J.* 2007;97:51-57.
8. Department of Health. *Report of the Second Phase of the Review of NHS Pathology Services in England*. London, UK: Department of Health; 2008.
9. Vreeman DJ, Finnell JT, Overhage JM. A rationale for parsimonious laboratory term mapping by frequency. *Proc AMIA Annu Fall Symp.* 2007;2007:771-775.
10. Sohn AJ, Hickner JM, Alem F. Use of point-of-care tests (POCTs) by US primary care physicians. *J Am Board Fam Med.* 2016;29:371-376.
11. Howick J, Cals JW, Jones C, et al. Current and future use of point-of-care tests in primary care: an international survey in Australia, Belgium, the Netherlands, the UK and the USA. *BMJ Open.* 2014;4:e005611.
12. Grand View Research. Point of care (POC) diagnostics/testing market size, share & trends analysis report by product (glucose testing, Hb1Ac, coagulation, fertility, cardiac markers, hematology, urinalysis), by end-use, and segment forecasts, 2018-2024. https://www.grandviewresearch.com/industry-analysis/point-of-care-poc-diagnostics-industry. Accessed July 20, 2018.
13. Allied Market Research. IVD market by product type—global opportunity analysis and industry forecast, 2017-2023. https://www.alliedmarketresearch.com/ivd-in-vitro-diagnostics-market. Accessed July 20, 2018.
14. Wilson ML, Fleming KA, Kuti MA, et al. Access to pathology and laboratory medicine services: a crucial gap. *Lancet.* 2018;391:1927-1938.
15. Sayed S, Cherniak W, Lawler M, et al. Improving pathology and laboratory medicine in low-income and middle-income countries: roadmap to solutions. *Lancet.* 2018;391:1939-1952.

Downloaded from academic.oup.com/ajcp/article/151/5/446/5237639 by University of Calgary user on 15 October 2026

© American Society for Clinical Pathology

DOI: 10.1093/ajcp/aqy165